

# Entropic Gravity and Temporal Gradients : $\Phi(x)$ as a Thermodynamic Vector

Benjamin Br  cheteau

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brecheteaub@gmail.com

## R  sum  

Gravity has long been described as geometry—the curvature of spacetime under the weight of energy. A complementary line of work, from Jacobson to Verlinde, suggests another view : gravity as thermodynamics, emerging from microscopic degrees of freedom and information flow. If space bears an entropic structure, then time must carry a gradient of entropy—a rhythm of becoming that regulates the flow of information.

We formalize this rhythm by the Chronon Field  $\Phi(x)$ , a local temporal rate with  $[\Phi] = \text{T}^{-1}$ , and by its thermodynamic 4-gradient  $\Phi_\mu \equiv \nabla_\mu \ln \Phi$ . In this framing,  $\nabla_\mu S$  aligns with  $\Phi_\mu$ , gravitational redshift traces  $\nabla \ln \Phi$ , and cosmic expansion follows  $H = -\dot{\Phi}/\Phi$ . We sketch an effective action for  $\Phi$ , exhibit a Newtonian limit where  $\mathbf{g} = -c^2 \nabla \ln \Phi$ , and state falsifiable predictions involving optical clock networks, relativistic geodesy, and large-scale structure ( $f\sigma_8$ ). Rhythm replaces curvature as the operative grammar : coherence, gravity, and the arrow of time become three faces of one temporal gradient.

**Keywords :** Entropic gravity, Chronon Field, temporal gradient, arrow of time, thermodynamics, quantum coherence, emergent gravity, public time (UTC/GNSS)

## ENTROPIC GRAVITY AND TEMPORAL GRADIENTS : $\Phi(x)$ AS A THERMODYNAMIC VECTOR

### I — From geometry to thermodynamics

In 1995, Ted Jacobson derived Einstein’s field equations from the first law of thermodynamics,

$$\delta Q = T dS,$$

applied to local Rindler horizons : spacetime curvature emerges as a balance between energy flux ( $\delta Q$ ) and entropy change ( $dS$ ). Later, Erik Verlinde argued that gravity may be an entropic tendency : a macroscopic manifestation of information striving toward equilibrium.

If gravity is thermodynamic in origin, time should be too : the arrow of time indicates the direction of entropy increase, and the *rate* of temporal passage should reflect the local rate of entropy production. The ticking of a clock, from atomic standards to cosmological redshift, is then a metronome of the universe’s drift toward disorder.

### II — The Chronon Field as temporal potential

We *postulate* that  $\Phi(x)$  encodes the local tempo of physical renewal (density of state transitions per unit time), so that spatiotemporal variations in  $\Phi$  track gradients of entropy production. By construction,  $[\Phi] = \text{s}^{-1}$ .

*Dimensions & conventions.* We work on a 4D Lorentzian manifold with metric signature  $(-, +, +, +)$ , SI units, and explicit  $c$ . Define the thermodynamic 4-gradient of the tempo

$$\Phi_\mu \equiv \nabla_\mu \ln \Phi,$$

and the entropy current  $J_S^\mu \equiv s u^\mu$ , with local entropy density  $s$  and 4-velocity  $u^\mu$ ; entropy production obeys

$$\nabla_\mu J_S^\mu = \sigma \geq 0.$$

Phenomenologically we posit a coarse-grained coupling on windows  $(\ell, \tau)$ ,

$$\nabla_\mu S \propto \Phi_\mu \quad \text{or equivalently} \quad \Phi_\mu \propto \frac{1}{T} \nabla_\mu s,$$

where  $T$  is the local temperature and the proportionality absorbs transport coefficients. These relations define  $\Phi_\mu$  as a *thermodynamic vector* : the direction and intensity of temporal ordering.

**Master relations (re-parameterizations).**

$$d\tau = \Phi^{-1}(x) dt \quad ; \quad a(t) \propto \Phi^{-1}(t) \quad ; \quad H = -\dot{\Phi}/\Phi \quad ; \quad J^\mu = \Phi u^\mu \quad ; \quad \nabla_\mu J^\mu = \Gamma.$$

They *relabel* durations and expansion without introducing new light cones or stress-energy.

*Lagrangian note (statutory).* Any effective action for  $\Phi$  is *non-energetic* : it generates kinematic re-parameterizations of clock rates but contributes no stress-energy component ; see the *Zero-Energy Lemma* (Art. 2).

**Operational residual (clocks).**

$$\left(\frac{\Delta f}{f}\right)_{\text{res}} \equiv \left(\frac{\Delta f}{f}\right)_{\text{obs}} - \left(\frac{\Delta f}{f}\right)_{\text{GR}} = \varepsilon_\Phi, \quad (1)$$

with target  $|\varepsilon_\Phi| \lesssim 10^{-3}$  of the GR signal on 1–10 m steps at  $10^{-18}$  stability (used in the geodesy box below).

### III — Gravity as a gradient of temporal order

General relativity states : matter tells spacetime how to curve. Rhythmic thermodynamics adds (as a *weak-field dictionary*) : entropy tells local clock rates how to vary.

A massive body concentrates energy and information, reconfiguring local entropy flux ; empirically, clocks run slower in deeper potentials. Identifying clock frequency with the local tempo,  $f \propto \Phi$ , the gravitational redshift law  $\Delta f/f \simeq \Delta\psi/c^2$  (with Newtonian potential  $\psi$ ) implies

$$\nabla \ln \Phi \simeq \frac{\nabla \psi}{c^2}.$$

Hence the observed acceleration in the weak-field limit

$$\mathbf{g} = -\nabla \psi \simeq -c^2 \nabla \ln \Phi,$$

so that free fall follows the slope of temporal tension.

*Statutory clarification (effective Newtonian law).*

$\mathbf{g} \simeq -c^2 \nabla \ln \Phi$

 valid for weak, quasi-stationary fields with  $|\Delta\Phi/\Phi| \ll 1$ , no new  $T_{\mu\nu}$ , null cones unchanged.

This is a phenomenological mapping at  $\mathcal{O}(c^{-2})$  ; GR's null cones and geodesics remain those of  $g_{\mu\nu}$  (see also Art. 6 for the cosmological relation  $H = -\dot{\Phi}/\Phi$ ).

**Encadré — Relativistic geodesy &  $\nabla\Phi$  (targets & null-tests).**

- **Vertical clocks (cm–m baselines).** Reference slope :  $\Delta f/f \simeq 1.1 \times 10^{-16} \text{ m}^{-1}$ . *Target* : bound any non-metric residual  $\varepsilon_\Phi$  after GR subtraction at the  $\lesssim 10^{-18}$  level, i.e.  $|\varepsilon_\Phi| \lesssim 10^{-3}$  of the gravitational signal on 1–10 m steps.

- **Long baselines** ( $10^2$ – $10^4$  km). Search for common-mode fractional-frequency noise  $S_y(f)$  consistent with regional  $\Phi$  fluctuations; *null-tests* : link independence (fiber/GNSS/free-space), A/B site permutations, environmental vetoes.
- **Acceptance criteria.** Persistence under link swap, invariance under duty-cycle permutations, dimensionless neutral witnesses (control oscillators), and cross-correlation confidence  $> 5\sigma$  after multiple-comparison correction.

#### IV — The arrow of time as rhythmic dissipation

Entropy increase defines the arrow of time; the Chronon Field adds heterogeneity to its *rate*. In high-entropy environments (supernovae, turbulent plasmas),  $\Phi$  accelerates and the arrow intensifies. In coherent phases (Bose–Einstein condensates, superconductors, quantum memories),  $\Phi$  is locally minimal (yet never vanishing), and the arrow softens.

The second law reads as phase alignment : systems evolve so that their local  $\Phi$  tends to synchronize with the surrounding field. Irreversibility is then a phenomenon of partial synchronization ; when coherence dominates, entropy production stalls and time’s beat attenuates.

#### V — Quantum coherence and thermal rhythm

In quantum systems, coherence is temporary resistance to entropy growth. Decoherence occurs when local oscillations desynchronize from  $\Phi$ . Across platforms, typical coherence times anti-correlate with environmental entropy rates :

solids (warm) : ps–ns ; superconducting circuits :  $\mu$ s–ms ; trapped ions : s–min.

This pattern is consistent with  $\Gamma(x) \propto \Phi(x)$  : the larger the local tempo of dissipation, the faster information loses phase.

*Indicative map (platform  $\rightarrow T_2$ ).*

| Platform                           | Typical environment    | $T_2$ scale |
|------------------------------------|------------------------|-------------|
| Warm solid-state (NV/quantum dots) | 300 K, phonons         | ps–ns       |
| Superconducting qubits             | mK, engineered baths   | $\mu$ s–ms  |
| Trapped ions / neutral atoms       | cryo/ultra-high vacuum | s–min       |

At one extreme, near-ideal coherence suppresses decoherence rates at millikelvin temperatures and  $\Phi$  is locally minimal ; at the other, maximal dissipation (e.g., Hawking emission) amplifies  $\Phi$  and temporal flow becomes violent. Between these limits, reality maintains itself by sustaining finite rhythm.

#### VI — Extreme regimes : black holes and caveats

For black holes, Bekenstein–Hawking entropy,

$$S_{\text{BH}} = \frac{k_B c^3 A}{4G\hbar} \simeq 1.05 \times 10^{77} k_B \left( \frac{M}{M_\odot} \right)^2,$$

and, when accretion is relevant, a parametric entropy rate

$$\frac{dS}{dt} = \frac{8\pi G k_B}{\hbar c} M \dot{M}$$

set upper intensities for  $\Phi$  in nature. The interpretation of *local* entropy production requires a specified control volume and foliation ; one should not identify “large  $\Phi$ ” with instantaneous loss of quantum coherence without stating the coarse-graining window  $(\ell, \tau)$ .

#### VII — Cosmological implications

On cosmic scales,  $\Phi(x)$  defines a global temporal potential whose secular decay shapes expansion. If the universal beat decelerates, then

$$H \equiv \frac{\dot{a}}{a} = - \frac{\dot{\Phi}}{\Phi}.$$

Redshift expresses the gradual loss of tempo ; today's  $T \simeq 2.7$  K marks the faint amplitude of the present rhythm. As a linking dictionary (phenomenological, weak-field),

$$\nabla_\mu S \leftrightarrow \Phi_\mu(x) \quad (\text{alignment}) \quad \text{while} \quad g_{\mu\nu} \text{ remains the GR geometry,}$$

so that entropy flow and temporal gradients co-vary without introducing new light-cones or stress-energy. Attraction, expansion, and decoherence appear as projections of the same pulse : the migration of entropy through time.

A convenient late-time rhythmic scale is  $\hbar H_0 \sim 10^{-33}$  eV (ultra-soft).

### VIII — Predictions & protocols (concise)

**P1 — Network clocks (homogeneous drift).** Long-baseline comparisons of optical clocks constrain a secular drift  $\dot{\Phi}/\Phi$ . Target :

$$|\dot{\Phi}/\Phi| \lesssim 10^{-19} \text{ s}^{-1},$$

via phase-locked arrays over 1–2 years with environmental decorrelation.

**P2 — Relativistic geodesy (spatial gradients).** Map  $\nabla \ln \Phi$  across altitude/temperature contrasts using portable optical clocks ; criterion : non-metric residuals after standard GR correction. (Reference slope as in the geodesy box, Sec. III.)

**P3 — Large-scale structure.** If  $H = -\dot{\Phi}/\Phi$  holds strictly, growth-rate observables  $f\sigma_8(z)$  show a predictable redshift dependence in the sign and magnitude of deviations from  $\Lambda$ CDM. Forecasts should marginalize over bias and neutrino mass, and report a two-parameter  $\{\Phi_0, \dot{\Phi}/\Phi\}$  contour, with *consistency checks against Art. 6 (CFC)* : use anisotropic BAO/AP (dimensionless  $F_{\text{AP}}$ ) and strong-lens  $D_{\Delta t}$  to break normalization degeneracies.

**P4 — Decoherence cartography.** Co-locate clock comparisons with qubit interferometry to test  $\Gamma \propto \Phi$  across controlled thermal/phononic gradients ; report joint bounds  $\{\dot{\Phi}/\Phi, \partial_i \ln \Phi, \xi\}$  with  $\xi$  the proportionality in  $\Gamma = \Gamma_{\text{env}} + \xi \Phi$ .

### IX — Box (entropic Newtonian limit, toy)

*Assumptions.* (i) weak, stationary field ; (ii) reference frequency  $f \propto \Phi$  ; (iii) small redshift  $\Delta f/f \simeq \Delta\psi/c^2$ .

*Four-line dictionary.*

$$\nabla \ln \Phi \simeq \frac{\nabla \psi}{c^2} \quad \Rightarrow \quad \mathbf{g} \equiv -\nabla \psi \simeq -c^2 \nabla \ln \Phi.$$

Thus the “gravitational force” is the slope of temporal tension  $\ln \Phi$ . If  $\Phi \simeq \Phi_0 e^{\psi/c^2}$ , then  $\psi$  recovers the Newtonian potential at  $\mathcal{O}(c^{-2})$ . (For the statutory form and domain of validity, see Eq. ?? in Sec. III.)

### X — Kinematic relaxation for $\Phi$ (non-energetic)

Rather than introducing a stress-energy for  $\Phi$ , we adopt a coarse-grained, *kinematic* relaxation law sourced by entropy production,

$$u^\mu \nabla_\mu \Phi = -\gamma (\Phi - \Phi_0) + D \Delta_\perp \Phi + \kappa \sigma,$$

where  $\gamma > 0$  is a small relaxation rate,  $D \geq 0$  an effective diffusivity (with  $\Delta_\perp \equiv h^{\mu\nu} \nabla_\mu \nabla_\nu$  and  $h^{\mu\nu} = g^{\mu\nu} + u^\mu u^\nu$ ), and  $\kappa$  a coupling to the entropy production density  $\sigma = \nabla_\mu J_S^\mu$ . On FLRW backgrounds this reduces to a first-order secular drift consistent with  $H = -\dot{\Phi}/\Phi$ . Crucially, this evolution *reparametrizes* clock rates while leaving  $g_{\mu\nu}$  and null cones governed by standard GR.

### XI — Conclusion : the rhythm of thermodynamics

The Chronon Field frames gravity, in the weak-field phenomenology, as modulation of temporal entropy rates. It binds the arrow of time to the architecture of spacetime : energy curves ; entropy conducts ; rhythm endures.

In  $\Phi(x)$ , the cosmos breathes between coherence and collapse. Gravity, heat, and time cease to be distinct ontologies and appear as synchronized beats of one music. The universe does not fall toward mass ; it flows along rhythm. What persists is less a geometry than a pulse by which everything keeps becoming.

This rhythmic thermodynamics extends the core thesis of *Time May Not Exist!* : the unity of time is *achieved*, not given, by local synchronization of  $\Phi(x)$  — the beat by which the universe keeps itself coherent.

